Chapter 5
CONFIDENCE INTERVALS

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CHAPTER OUTLINE

Statistical Intervals for a Single Sample

1. CONFIDENCE INTERVAL ON THE MEAN OF A NORMAL DISTRIBUTION, VARIANCE KNOWN
   - Development of the Confidence Interval and Its Basic Properties
   - Choice of Sample Size
   - One-Sided Confidence Bounds
   - Large-Sample Confidence Interval for μ
2. CONFIDENCE INTERVAL ON THE MEAN OF A NORMAL DISTRIBUTION, VARIANCE UNKNOWN
   - t Distribution
   - t Confidence Interval on μ
3. CONFIDENCE INTERVAL ON THE VARIANCE AND STANDARD DEVIATION OF A NORMAL DISTRIBUTION
4. LARGE-SAMPLE CONFIDENCE INTERVAL FOR A POPULATION PROPORTION

LEARNING OBJECTIVES

Learning Objectives

After careful study of this chapter you should be able to do the following:

1. Construct confidence intervals on the mean of a normal distribution, using either the normal distribution or the t-distribution method
2. Construct confidence intervals on the variance and standard deviation of a normal distribution
3. Construct confidence intervals on a population proportion

INTRODUCTION

Interval estimate

An interval estimate of a population parameter θ is an interval of the form $\widehat{\theta}_L < \theta < \widehat{\theta}_U$, where $\widehat{\theta}_L$ and $\widehat{\theta}_U$ depend on the value of the statistic $\widehat{\Theta}$ for a particular sample and also on the sampling distribution of $\widehat{\Theta}$.

Interpretation of Interval Estimates

- From the sampling distribution of $\widehat{\Theta}$ we shall be able to determine $\widehat{\Theta}_L$ and $\widehat{\Theta}_U$ such that $P(\widehat{\Theta}_L < \theta < \widehat{\Theta}_U)$ is equal to any positive fractional value we care to specify. If, for instance, we find $\widehat{\Theta}_L$ and $\widehat{\Theta}_U$ such that

$$\boxed{P(\widehat{\Theta}_L < \theta < \widehat{\Theta}_U) = 1 - \alpha} \tag{1}$$

  for $0 < \alpha < 1$, then we have a probability of $1 - \alpha$ of selecting a random sample that will produce an interval containing θ.
- The interval $\widehat{\theta}_L < \theta < \widehat{\theta}_U$, computed from the selected sample, is called a confidence interval.

INTRODUCTION

Interpretation of Interval Estimates

- The interval $\widehat{\theta}_L < \theta < \widehat{\theta}_U$, computed from the selected sample, is called a $100(1-\alpha)\%$ confidence interval.
- The endpoints, $\widehat{\theta}_L$ and $\widehat{\theta}_U$, are called the lower and upper confidence limits.
- When $\alpha = 0.05$, we have a 95% confidence interval, and when $\alpha = 0.01$, we obtain a wider 99% confidence interval.
- The wider the confidence interval is $\widehat{\theta}_U - \widehat{\theta}_L$, the more confident we can be that the interval contains the unknown parameter.

Content

Development of the Confidence Interval and Its Basic Properties

Introduction

- The basic ideas of a confidence interval (CI) are most easily understood by initially considering a simple situation.
- Suppose that we have a normal population with unknown mean μ and known variance σ^2.
- This is a somewhat unrealistic scenario because typically both the mean and variance are unknown.
- However, in subsequent sections we will present confidence intervals for more general situations.

Development of the Confidence Interval and Its Basic Properties

Development of the Confidence Interval

- Suppose that $(X_1, X_2, \ldots, X_n)$ is a random sample from a normal distribution with unknown mean μ and known variance σ^2. From the results of Chapter 4 we know that the sample mean $\overline{X}$ is normally distributed with mean μ and variance σ^2/n. We may standardize $\overline{X}$ by subtracting the mean and dividing by the standard deviation, which results in the variable

$$\boxed{Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}} \tag{2}$$

  The random variable Z has a standard normal distribution.

- Because Z in (2) has a standard normal distribution, we may write

$$P\Big(-z_{\alpha/2} < \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\Big) = 1 - \alpha,$$

  where $z_{\alpha/2}$ is the z-value which we find an area of $\alpha/2$ under the normal curve (see Figure 1).

Development of the Confidence Interval and Its Basic Properties

Development of the Confidence Interval

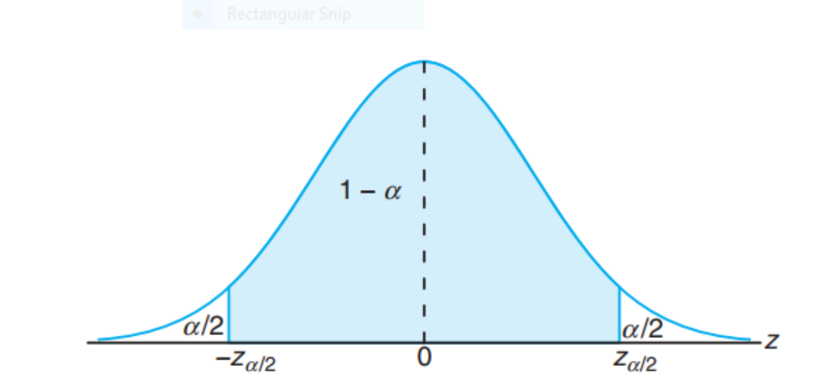

Figure 1: $P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$

Development of the Confidence Interval and Its Basic Properties

Development of the Confidence Interval

- So,

$$\boxed{P\left(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha} \tag{3}$$

- From consideration of Equation (3), the lower and upper limits of the inequalities in Equation (3) are the lower- and upper-confidence limits $\widehat{\theta}_L$ and $\widehat{\theta}_U$, respectively. This leads to the following definition.

Theorem 1 (Confidence Interval on μ, σ^2 Known)

If $\overline{x}$ is the mean of a random sample of size n from a population with known variance σ^2, a $100(1-\alpha)\%$ confidence interval for μ is given by

$$\boxed{\overline{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}} \tag{4}$$

where $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right.

Development of the Confidence Interval and Its Basic Properties

Example 1

The average zinc concentration recovered from a sample of measurements taken in 36 different locations in a river is found to be 2.6 grams per milliliter. Find the 95% and 99% confidence intervals for the mean zinc concentration in the river. Assume that the zinc concentration in the river is normally distributed with standard deviation is 0.3 gram per milliliter.

Solution

- The point estimate of μ is $\overline{x} = 2.6$.
- The z-value leaving an area of 0.025 to the right, and therefore an area of 0.975 to the left, is $z_{0.025} = 1.96$.
- Hence, the 95% confidence interval is

$$2.6 - (1.96)\left(\frac{0.3}{\sqrt{36}}\right) < \mu < 2.6 + (1.96)\left(\frac{0.3}{\sqrt{36}}\right)$$

which reduces to $2.502 < \mu < 2.698$.

Development of the Confidence Interval and Its Basic Properties

Solution (continuous)

- To find a 99% confidence interval, we find the z-value leaving an area of 0.005 to the right and 0.995 to the left.
- We have $z_{0.005} = 2.58$, and the 99% confidence interval is

$$2.6 - (2.58)\left(\frac{0.3}{\sqrt{36}}\right) < \mu < 2.6 + (2.58)\left(\frac{0.3}{\sqrt{36}}\right)$$

or simply

$$2.471 < \mu < 2.729.$$

Table

Table 1 $\left(\Phi(x) = \dfrac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{\frac{-t^2}{2}} dt\right)$

x	0	1	2	3	4	5	6	7	8	9
0.0	0.50000	50399	50798	51197	51595	51994	52392	52790	53188	53586
0.1	53983	54380	54776	55172	55567	55962	56356	56749	57142	57535
0.2	57926	58317	58706	59095	59483	59871	60257	60642	61026	61409
0.3	61791	62172	62556	62930	63307	63683	64058	64431	64803	65173
0.4	65542	65910	66276	66640	67003	67364	67724	68082	68439	68739
0.5	69146	69447	69847	70194	70544	70884	71226	71566	71904	72240
0.6	72575	72907	73237	73565	73891	74215	74537	74857	75175	75490
0.7	75804	76115	76424	76730	77035	77337	77637	77935	78230	78524
0.8	78814	79103	79389	79673	79955	80234	80511	80785	81057	81327
0.9	81594	81859	82121	82381	82639	82894	83147	83398	83646	83891
1.0	84134	84375	84614	84850	85083	85314	85543	85769	85993	86214
1.1	86433	86650	86864	87076	87286	87493	87698	87900	88100	88298
1.2	88493	88686	88877	89065	89251	89435	89617	89796	89973	90147
1.3	90320	90490	90658	90824	90988	91149	91309	91466	91621	91774
1.4	91924	92073	92220	92364	92507	92647	92786	92922	93056	93189
1.5	93319	93448	93574	93699	93822	93943	94062	94179	94295	94408
1.6	94520	94630	94738	94845	94950	95053	95154	95254	95352	95449
1.7	95543	95637	95728	95818	95907	95994	96080	96164	96246	96327
1.8	96407	96485	96562	96638	96712	96784	96856	96926	96995	97062
1.9	97128	97193	97257	97320	97381	97441	97500	97558	97615	97670

Table 1 (continuous)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{\frac{-t^2}{2}}\,dt$$

x	0	1	2	3	4	5	6	7	8	9
2.0	97725	97778	97831	97882	97932	97982	98030	98077	98124	98169
2.1	98214	98257	98300	98341	98382	98422	99461	98500	98537	98574
2.2	98610	98645	98679	98713	98745	98778	98809	98840	98870	98899
2.3	98928	98956	98983	99010	99036	99061	99086	99111	99134	99158
2.4	99180	99202	99224	99245	99266	99285	99305	99324	99343	99361
2.5	99379	99396	99413	99430	99446	99261	99477	99492	99506	99520
2.6	99534	99547	99560	99573	99585	99598	99609	99621	99632	99643
2.7	99653	99664	99674	99683	99693	99702	99711	99720	99728	99763
2.8	99744	99752	99760	99767	99774	99781	99788	99795	99801	99807
2.9	99813	99819	99825	99831	99836	99841	99846	99851	99856	99861
3.0	0,99865	3,1	99903	3,2	99931	3,3	99952	3,4	99966	
3.5	99977	3,6	99984	3,7	99989	3,8	99993	3,9	99995	
4.0	999968									
4.5	999997									
5.0	99999997									

Choice of Sample Size

Error

- The precision of the confidence interval in Equation (4) is $2z_{\alpha/2}\sigma/\sqrt{n}$. This means that in using $\overline{x}$ to estimate μ, the error $E = |\overline{x} - \mu|$ is less than or equal to $z_{\alpha/2}\sigma/\sqrt{n}$ with confidence $100(1-\alpha)$.
- In situations where the sample size can be controlled, we can choose n so that we are $100(1-\alpha)$ percent confident that the error in estimating μ is less than a specified bound on the error E.
- The appropriate sample size is found by choosing n such that $z_{\alpha/2}\sigma/\sqrt{n} = E$. Solving this equation gives the following formula for n.

Theorem 2 (Sample Size for Specified Error on the Mean, Variance Known)

If $\overline{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\overline{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$\boxed{n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2} \tag{5}$$

Choice of Sample Size

Note

- If the right-hand side of Equation (5) is not an integer, it must be rounded up. This will ensure that the level of confidence does not fall below $100(1-\alpha)\%$.
- Notice that $2E$ is the length of the resulting confidence interval.

Example 2

In Example 1, we are

1. 95% confident that the sample mean $\overline{x} = 2.6$ differs from the true mean μ by an amount less than $E = (1.96)(0.3)/\sqrt{36} = 0.098$ and
2. 99% confident that the difference is less than $E = (2.58)(0.3)/\sqrt{36} = 0.129$.

Choice of Sample Size

Example 3

How large a sample is required if we want to be 95% confident that our estimate of μ in Example 1 is off by less than 0.05?

Solution

The population standard deviation is $\sigma = 0.3$. Then, by Theorem 2,

$$n = \left(\frac{(1.96)(0.3)}{0.05}\right)^2 = 138.2976$$

Therefore, we can be 95% confident that a random sample of size 139 will provide an estimate $\overline{x}$ differing from μ by an amount less than 0.05.

Choice of Sample Size

Remark

Notice the general relationship between sample size, desired length of the confidence interval $2E$, confidence level $100(1-\alpha)$, and standard deviation σ:

- As the desired length of the interval $2E$ decreases, the required sample size n increases for a fixed value of σ and specified confidence.
- As σ increases, the required sample size n increases for a fixed desired length $2E$ and specified confidence.
- As the level of confidence increases, the required sample size n increases for fixed desired length $2E$ and standard deviation σ.

One-Sided Confidence Bounds

Introduction

The confidence interval in Equation (4) gives both a lower confidence bound and an upper confidence bound for μ. Thus, it provides a two-sided CI. It is also possible to obtain one-sided confidence bounds for μ by setting either $\widehat{\theta}_L = -\infty$ or $\widehat{\theta}_U = +\infty$ and replacing $z_{\alpha/2}$ by z_α.

- From the Central Limit Theorem: $P\Big(\dfrac{\overline{X}-\mu}{\sigma/\sqrt{n}} < z_\alpha\Big) = 1-\alpha.$

  One can then manipulate the probability statement much as before and obtain

  $$P(\mu > \overline{X} - z_\alpha \sigma/\sqrt{n}) = 1-\alpha.$$

- Similar manipulation of $P\Big(\dfrac{\overline{X}-\mu}{\sigma/\sqrt{n}} > -z_\alpha\Big) = 1-\alpha$ gives

  $$P(\mu < \overline{X} + z_\alpha \sigma/\sqrt{n}) = 1-\alpha.$$

One-Sided Confidence Bounds

Theorem 3 (One-Sided Confidence Bounds on the Mean, Variance Known)

Let $\overline{x}$ be the mean of a random sample of size n from a population with variance σ^2.

1. A $100(1-\alpha)\%$ upper-confidence bounds for μ is

$$\boxed{\mu < \overline{x} + z_\alpha \sigma/\sqrt{n}} \tag{6}$$

2. A $100(1-\alpha)\%$ lower-confidence bounds for μ is

$$\boxed{\overline{x} - z_\alpha \sigma/\sqrt{n} < \mu} \tag{7}$$

.

One-Sided Confidence Bounds

Example 4

The same data for impact testing from Example 1 are used to construct a lower, one-sided 95% confidence interval for the mean zinc concentration in the river. Recall that $\overline{x} = 2.6$, $\sigma = 0.3$, and $n = 36$. The interval is

$$\overline{x} - z_\alpha \sigma/\sqrt{n} < \mu; \quad 2.6 - (1.645)(0.3)/\sqrt{36} < \mu; \quad 2.5178 < \mu \text{ (grams per milliliter)}.$$

Hence, we are 95% confident that the mean zinc concentration in the river is more than 2.5178 grams per milliliter.

Remark

The lower limit for the two-sided interval in Example 1 was 2.502. Because $z_\alpha < z_{\alpha/2}$, the lower limit of a one-sided interval is always greater than the lower limit of a two-sided interval of equal confidence. The onesided interval does not bound μ from above so that it still achieves 95% confidence with a slightly greater lower limit. If our interest is only in the lower limit for μ, then the one-sided interval is preferred because it provides equal confidence with a greater lower limit. Similarly, a one-sided upper limit is always less than a two-sided upper limit of equal confidence.

One-Sided Confidence Bounds

Example 5

In a psychological testing experiment, 25 subjects are selected randomly and their reaction time, in seconds, to a particular stimulus is measured. Past experience suggests that the variance in reaction times to these types of stimuli is 4 $\sec^2$ and that the distribution of reaction times is approximately normal. The average time for the subjects is 6.2 seconds. Give an upper 95% bound for the mean reaction time.

Solution

The upper 95% bound is given by

$$\overline{x} + z_\alpha \sigma/\sqrt{n} = 6.2 + (1.645)\sqrt{4/25} = 6.2 + 0.658 = 6.858 \text{ seconds.}$$

Hence, we are 95% confident that the mean reaction time is less than 6.858 seconds.

Large-Sample Confidence Interval for μ

Introduction

- We have assumed that the population distribution is normal with unknown mean and known standard deviation σ.
- We now present a large-sample CI for that does not require these assumptions.
- Let $(X_1, X_2, \ldots, X_n)$ be a random sample from a population with unknown mean μ and variance σ^2.
- Now if the sample size n is large, the central limit theorem implies that $\overline{X}$ has approximately a normal distribution with mean μ and variance σ^2/n.
- Therefore, $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$ has approximately a standard normal distribution. This ratio could be used as a pivotal quantity and manipulated as in previous section to produce an approximate CI for μ.
- However, the standard deviation σ is unknown. It turns out that when n is large, replacing σ by the sample standard deviation S has little effect on the distribution of Z.

Large-Sample Confidence Interval for μ

Theorem 4 (Large-Sample Confidence Interval on the Mean)

When n is large, the quantity

$$\frac{\overline{X}-\mu}{S/\sqrt{n}}$$

has an approximate standard normal distribution. Consequently,

$$\boxed{\overline{x}-z_{\alpha/2}\frac{s}{\sqrt{n}}<\mu<\overline{x}+z_{\alpha/2}\frac{s}{\sqrt{n}}} \tag{8}$$

is a large sample confidence interval for μ, with confidence level of approximately $100(1-\alpha)\%$.

Note

Equation (8) holds regardless of the shape of the population distribution. Generally n should be at least 40 to use this result reliably. The central limit theorem generally holds for $n \geq 30$, but the larger sample size is recommended here because replacing σ by S in Z results in additional variability.

Large-Sample Confidence Interval for μ

Example 6

Scholastic Aptitude Test (SAT) mathematics scores of a random sample of 500 high school seniors in the state of Texas are collected, and the sample mean and standard deviation are found to be 501 and 112, respectively. Find a 99% confidence interval on the mean SAT mathematics score for seniors in the state of Texas.

Solution

Since the sample size is large, it is reasonable to use the normal approximation. Using Table 1, we find $z_{0.005} = 2.575$. Hence, a 99% confidence interval for μ is

$$501 - (2.575)\left(\frac{112}{\sqrt{500}}\right) < \mu < 501 + (2.575)\left(\frac{112}{\sqrt{500}}\right),$$

which yields $488.1023 < \mu < 513.8976$.

Content

Introduction

Introduction

- Suppose that the population of interest has a normal distribution with unknown mean μ and unknown variance σ^2. Assume that a random sample of size n, say, $(X_1, X_2, \ldots, X_n)$, is available, and let $\overline{X}$ and S^2 be the sample mean and variance, respectively.
- We wish to construct a two-sided CI on μ. If the variance σ^2 is known, we know that $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$ has a standard normal distribution.
- When σ^2 is unknown, a logical procedure is to replace σ with the sample standard deviation S. The random variable Z now becomes $T = (\overline{X} - \mu)/(S/\sqrt{n})$. A logical question is, what effect does replacing σ by S have on the distribution of the random variable T? If n is large, the answer to this question is "very little," and we can proceed to use the confidence interval based on the normal distribution from previous section. However, n is usually small in most engineering problems, and in this situation a different distribution must be employed to construct the CI.

t Distribution

Definition 1 (t Distribution)

Let $(X_1, X_2, \ldots, X_n)$ be a random sample from a normal distribution with unknown mean μ and unknown variance σ^2. The random variable

$$\boxed{T = \frac{\overline{X} - \mu}{S/\sqrt{n}}} \tag{9}$$

has a t distribution with $n-1$ degrees of freedom.

t-Distribution

Table 2

d.f. \ α	0.10	0.05	0.025	0.01	0.005	0.002	0.0005
1	3.078	6.314	12.706	31.821	63.526	318.309	363.6
2	1.886	2.920	4.303	6.965	9.925	22.327	31.600
3	1.638	2.353	3.128	4.541	5.841	10.215	12.922
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.705	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.606	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850

t-Distribution

Table 2 (continuous)

d.f. \ α	0.10	0.05	0.025	0.01	0.005	0.002	0.0005
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.796	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
$+\infty$	1.282	1.645	1.960	2.326	2.576	3.090	3.291

t Confidence Interval on μ

Confidence Interval

- It is easy to find a $100(1-\alpha)$ percent confidence interval on the mean of a normal distribution with unknown variance by proceeding essentially as we did in previous section. We know that the distribution of T in (9) is t with $n-1$ degrees of freedom.
- Letting $t_{\alpha/2}$ be the upper $100\alpha/2$ percentage point of the t distribution with $n-1$ degrees of freedom, we may write (see Figure 2)

$$P(-t_{\alpha/2} < T < t_{\alpha/2}) = 1-\alpha,$$

or

$$P\Big(-t_{\alpha/2} < \frac{\overline{X}-\mu}{S/\sqrt{n}} < t_{\alpha/2}\Big) = 1-\alpha.$$

- Hence

$$\boxed{P\Big(\overline{X} - t_{\alpha/2}\frac{S}{\sqrt{n}} < \mu < \overline{X} + t_{\alpha/2}\frac{S}{\sqrt{n}}\Big) = 1-\alpha} \tag{10}$$

t Confidence Interval on μ

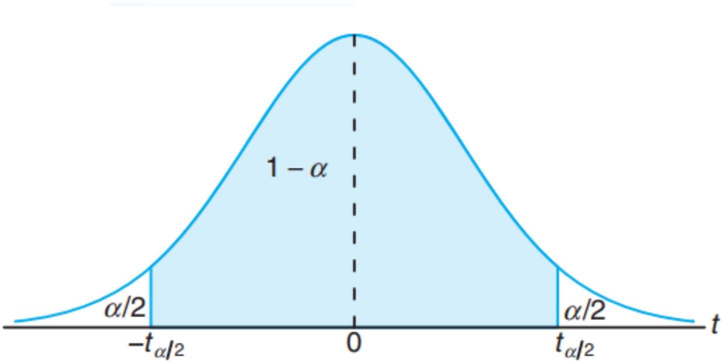

Figure 2: $P(-t_{\alpha/2} < T < t_{\alpha/2}) = 1 - \alpha,$

t Confidence Interval on μ

Theorem 5 (Confidence Interval on the Mean, Variance Unknown)

If $\overline{x}$ and s are the mean and standard deviation of a random sample from a normal population with unknown variance σ^2, a $100(1-\alpha)\%$ confidence interval on μ is given by

$$\boxed{\overline{x} - t_{\alpha/2}^{(n-1)} \frac{s}{\sqrt{n}} < \mu < \overline{x} + t_{\alpha/2}^{(n-1)} \frac{s}{\sqrt{n}}} \tag{11}$$

where $t_{\alpha/2}^{(n-1)}$ is the t-value with $\nu = n-1$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

t Confidence Interval on μ

One-Sided Confidence Bounds

One-sided confidence bounds on the mean of a normal distribution are also of interest and are easy to find. Simply use only the appropriate lower or upper confidence limit from Equation (11) and replace $t_{\alpha/2}^{(n-1)}$ by $t_{\alpha}^{(n-1)}$.

Theorem 6 (One-Sided Confidence Bounds on μ, σ^2 Unknown)

Computed one-sided confidence bounds for μ with σ unknown are as the reader would expect, namely

$$\text{Upper one-sided bound:} \quad \mu < \overline{x} + t_{\alpha}^{(n-1)} \frac{s}{\sqrt{n}}; \tag{12}$$

$$\text{Lower one-sided bound:} \quad \overline{x} - t_{\alpha}^{(n-1)} \frac{s}{\sqrt{n}} < \mu. \tag{13}$$

They are the upper and lower $100(1-\alpha)\%$ bounds, respectively. Here t_α is the t-value having an area of α to the right.

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t Confidence Interval on μ

Example 7

The contents of seven similar containers of sulfuric acid are 9.8, 10.2, 10.4, 9.8, 10.0, 10.2, and 9.6 liters. Find a 95% confidence interval for the mean contents of all such containers, assuming an approximately normal distribution.

Solution

- The sample mean and standard deviation for the given data are $\overline{x} = 10.0$ and $s = 0.283$.
- Using Table 2, we find $t_{0.025}^{(6)} = 2.447$ for $\nu = n - 1 = 6$ degrees of freedom.
- Hence, the 95% confidence interval for μ is

$$10.0 - (2.447)\left(\frac{0.283}{\sqrt{7}}\right) < \mu < 10.0 + (2.447)\left(\frac{0.283}{\sqrt{7}}\right),$$

which reduces to $9.7383 < \mu < 10.2617$.

Content

χ^2 Distribution

Introduction

Sometimes confidence intervals on the population variance or standard deviation are needed. When the population is modeled by a normal distribution, the intervals described in this section are applicable.

Theorem 7 (χ^2 Distribution)

Let $(X_1, X_2, \dots, X_n)$ be a random sample from a normal distribution with mean μ and variance σ^2, and let S^2 be the sample variance. Then the random variable

$$\boxed{\chi^2 = \frac{(n-1)S^2}{\sigma^2}} \tag{14}$$

has a chi-square χ^2 distribution with $n-1$ degrees of freedom.

χ^2 Distribution

χ^2 Distribution

- The probability density function of a χ^2 random variable is

$$\boxed{f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{(k/2)-1} e^{-e/2} \quad x > 0} \tag{15}$$

  where k is the number of degrees of freedom. The mean and variance of the χ^2 distribution are k and $2k$, respectively.

- Note that the chi-square random variable is nonnegative and that the probability distribution is skewed to the right. However, as k increases, the distribution becomes more symmetric. As $k \to \infty$, the limiting form of the chi-square distribution is the normal distribution (see Chapter 4).
- The percentage points of the χ^2 distribution are given in Table 3.

χ^2 Distribution

χ^2 Distribution

- Define $\chi^2_{\alpha,k}$ as the percentage point or value of the chi-square random variable with k degrees of freedom such that the probability that χ^2 exceeds this value is α. That is,

$$P(\chi^2 > \chi^2_{\alpha,k}) = \int_{\chi^2_{\alpha,k}}^{\infty} f(u)du = \alpha$$

  This probability is shown as the shaded area in Fig. 3.

- To illustrate the use of Table 3, note that the areas α are the column headings and the degrees of freedom k are given in the left column. Therefore, the value with 10 degrees of freedom having an area (probability) of 0.05 to the right is $\chi^2_{0.05,10} = 18.31$. This value is often called an upper 5% point of chi-square with 10 degrees of freedom. We may write this as a probability statement as follows:

$$P(\chi^2 > \chi^2_{0.05,10}) = P(\chi^2 > 18.31) = 0.05.$$

  Conversely, a lower 5% point of chi-square with 10 degrees of freedom would be $\chi^2_{0.95,10} = 3.94$. Both of these percentage points are shown in Figure 4.

χ^2 Distribution

χ^2 Distribution

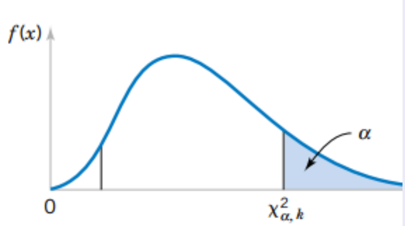

Figure 3: The percentage point $\chi^2_{\alpha,k}$

χ^2 Distribution

χ^2 Distribution

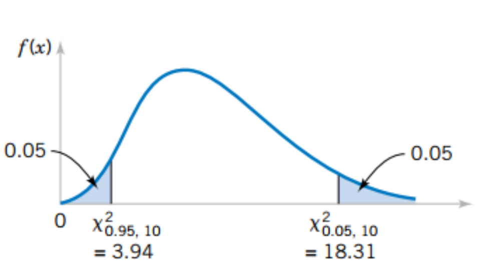

Figure 4: The upper percentage point $\chi^2_{0.05,10} = 18.31$ and the lower percentage point $\chi^2_{0.95,10} = 3.94$

χ^2 Distribution

Table 3 (χ^2 Distribution)

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	8.55	14.34	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	15.34	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.87	17.34	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	18.34	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57	40.00

χ^2 Distribution

Table 3 (continuous)

19	6.84	7.63	8.91	10.12	11.65	18.34	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	24.34	34.28	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96	49.65
28	12.46	13.57	15.31	16.93	18.94	27.34	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	28.34	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	29.34	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	39.34	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	49.33	63.17	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	46.46	59.33	74.40	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	55.33	69.33	85.53	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	73.29	89.33	107.57	113.14	118.14	124.12	128.30
100	67.33	70.06	74.22	77.93	82.36	99.33	118.50	124.34	129.56	135.81	140.17

ν = degrees of freedom.

Confidence Interval on the Variance

Confidence Interval on the Variance

- The construction of the $100(1-\alpha)\%$ CI for σ^2 is straightforward. Because

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

is chi-square with $n-1$ degrees of freedom, we may write

$$P(\chi^2_{1-\alpha/2,n-1} < \chi^2 < \chi^2_{\alpha/2,n-1}) = 1-\alpha.$$

- So that

$$P\left(\chi^2_{1-\alpha/2,n-1} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{\alpha/2,n-1}\right) = 1-\alpha.$$

- This last equation can be rearranged as

$$\boxed{P\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right) = 1-\alpha} \tag{16}$$

Confidence Interval on the Variance

Theorem 8 (Confidence Interval on the Variance)

If s^2 is the sample variance from a random sample of n observations from a normal distribution with unknown variance σ^2, then a $100(1-\alpha)\%$ confidence interval on σ^2 is

$$\boxed{\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}} \tag{17}$$

where $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ are the upper and lower $100\alpha/2$ percentage points of the chi-square distribution with $n-1$ degrees of freedom, respectively.

Confidence Interval on the Standard Deviation

A confidence interval for σ has lower and upper limits that are the square roots of the corresponding limits in Equation (17).

One-Sided Confidence Bounds on the Variance

Theorem 9 (One-Sided Confidence Bounds on the Variance)

The $100(1-\alpha)\%$ lower and upper confidence bounds on σ^2 are

$$\boxed{\frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} < \sigma^2 \quad \text{and} \quad \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}}} \tag{18}$$

respectively.

One-Sided Confidence Bounds on the Variance

Example 8

An automatic filling machine is used to fill bottles with liquid detergent. A random sample of 20 bottles results in a sample variance of fill volume of $s^2 = 0.0153$ (fluid ounce)^2. If the variance of fill volume is too large, an unacceptable proportion of bottles will be under- or overfilled. We will assume that the fill volume is approximately normally distributed. A 95% upper confidence bound is found from Equation (18) as follows:

$$\sigma^2 < \frac{(n-1)s^2}{\chi^2_{0.95,19}} \quad \text{or} \quad \sigma^2 < \frac{(19)(0.0153)}{10.117} = 0.0287 \text{ (fluid ounce)}^2.$$

This last expression may be converted into a confidence interval on the standard deviation σ by taking the square root of both sides, resulting in

$$\sigma < 0.1694.$$

Practical Interpretation: Therefore, at the 95% level of confidence, the data indicate that the process standard deviation could be as large as 0.17 fluid ounce. The process engineer or manager now needs to determine if a standard deviation this large could lead to an operational problem with under-or over filled bottles.

Content

Normal Approximation for a Binomial Proportion

Introduction

- It is often necessary to construct confidence intervals on a population proportion. For example, suppose that a random sample of size n has been taken from a large (possibly infinite) population and that $X(\leq n)$ observations in this sample belong to a class of interest. Then $\widehat{P} = X/n$ is a point estimator of the proportion of the population p that belongs to this class.
- Note that n and p are the parameters of a binomial distribution. Furthermore, By the Central Limit Theorem, $\hat{P}$ is approximately normally distributed with mean

$$\mu_{\widehat{P}} = E(\widehat{P}) = E\left(\frac{X}{n}\right) = \frac{np}{n} = p$$

and variance

$$\sigma^2_{\widehat{P}} = \sigma^2_{X/n} = \frac{\sigma^2_X}{n^2} = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n},$$

if p is not too close to either 0 or 1 and if n is relatively large.
- Typically, to apply this approximation we require that np and $n(1-p)$ be greater than or equal to 5.

Normal Approximation for a Binomial Proportion

Theorem 10 (Normal Approximation for a Binomial Proportion)

If n is large, the distribution of

$$\boxed{Z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{\widehat{P} - p}{\sqrt{\frac{p(1-p)}{n}}}} \tag{19}$$

is approximately standard normal.

Normal Approximation for a Binomial Proportion

Construct the Confidence Interval

- To construct the confidence interval on p, note that

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) \simeq 1-\alpha$$

- Substituting for Z, we write

$$P\left(-z_{\alpha/2} < \frac{\widehat{P}-p}{\sqrt{p(1-p)/n}} < z_{\alpha/2}\right) \simeq 1-\alpha.$$

- This may be rearranged as

$$\boxed{P\left(\widehat{P} - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} < p < \widehat{P} + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right) \simeq 1-\alpha} \tag{20}$$

Normal Approximation for a Binomial Proportion

Construct the Confidence Interval

- The quantity $\sqrt{p(1-p)/n}$ in Equation (20) is called the standard error of the point estimator. Unfortunately, the upper and lower limits of the confidence interval obtained from Equation (20) contain the unknown parameter p. However, as suggested at the end of previous section, a satisfactory solution is to replace p by $\widehat{P}$ in the standard error, which results in

$$\boxed{P\left(\widehat{P} - z_{\alpha/2}\sqrt{\frac{\widehat{P}(1-\widehat{P})}{n}} < p < \widehat{P} + z_{\alpha/2}\sqrt{\frac{\widehat{P}(1-\widehat{P})}{n}}\right) \simeq 1-\alpha} \qquad (21)$$

This leads to the approximate $100(1-\alpha)\%$ confidence interval on p.

Normal Approximation for a Binomial Proportion

Theorem 11 (Approximate Confidence Interval on a Binomial Proportion)

If $\widehat{p}$ is the proportion of observations in a random sample of size n that belongs to a class of interest, an approximate $100(1-\alpha)\%$ confidence interval on the proportion p of the population that belongs to this class is

$$\boxed{\widehat{p} - z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} < p < \widehat{p} + z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}} \tag{22}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

Normal Approximation for a Binomial Proportion

Note

- This procedure depends on the adequacy of the normal approximation to the binomial. To be reasonably conservative, this requires that np and $n(1-p)$ be greater than or equal to 5. In situations where this approximation is inappropriate, particularly in cases where n is small, other methods must be used.
- To be on the safe side, one should require both $n\widehat{p}$ and $n(1-\widehat{p})$ to be greater than or equal to 5.

Example 9

In a random sample of $n = 500$ families owning television sets in the city of Hamilton, Canada, it is found that $x = 340$ subscribe to HBO. Find a 95% confidence interval for the actual proportion of families with television sets in this city that subscribe to HBO.

Normal Approximation for a Binomial Proportion

Solution

- The point estimate of p is $\widehat{p} = 340/500 = 0.68$. We find $z_{0.025} = 1.96$.
- Therefore, using (22), the 95% confidence interval for p is

$$0.68 - 1.96\sqrt{\frac{(0.68)(0.32)}{500}} < p < 0.68 + 1.96\sqrt{\frac{(0.68)(0.32)}{500}},$$

which simplifies to $0.6391 < p < 0.7209$.

Choice of Sample Size

Error

- Since $\widehat{P}$ is the point estimator of p, we can define the error in estimating p by $\widehat{P}$ as $|\widehat{P} - p|$. Note that we are approximately $100(1-\alpha)\%$ confident that this error is less than $z_{\alpha/2}\sqrt{\widehat{p}(1-\widehat{p})/n}$.
- For instance, in Example 9, we are 95% confident that the sample proportion $\widehat{p} = 0.68$ differs from the true proportion p by an amount not exceeding 0.04.

Theorem 12 (Error)

If $\widehat{p}$ is used as an estimate of p, we can be $100(1-\alpha)\%$ confident that the error will not exceed

$$\boxed{E = z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}} \tag{23}$$

Choice of Sample Size

Theorem 13 (Sample Size for a Specified Error on a Binomial Proportion)

If $\widehat{p}$ is used as an estimate of p, we can be $100(1-\alpha)\%$ confident that the error will be less than a specified amount E when the sample size is approximately

$$\boxed{n = \frac{z^2_{\alpha/2}\widehat{p}(1-\widehat{p})}{E^2}} \qquad (24)$$

Note

- An estimate of p is required to use Equation (24). If an estimate $\widehat{p}$ from a previous sample is available, it can be substituted for p in Equation (24), or perhaps a subjective estimate can be made. If these alternatives are unsatisfactory, a preliminary sample can be taken, computed, and then Equation (24) used to determine how many additional observations are required to estimate p with the desired accuracy.
- Another approach to choosing n uses the fact that the sample size from Equation (24) will always be a maximum for $\widehat{p}(1-\widehat{p}) = 0.25$, and this can be used to obtain an upper bound on n.

Choice of Sample Size

Theorem 14 (Choice of Sample Size)

If $\widehat{p}$ is used as an estimate of p, we can be at least $100(1-\alpha)\%$ confident that the error will not exceed a specified amount E when the sample size is

$$\boxed{n = \frac{z_{\alpha/2}^2}{4E^2}} \tag{25}$$

Choice of Sample Size

Example 10

How large a sample is required if we want to be 95% confident that our estimate of p in Example 9 is within 0.02 of the true value?

Solution

Let us treat the 500 families as a preliminary sample, providing an estimate $\widehat{p} = 0.68$. Then, by Theorem 13,

$$n = \frac{(1.96)^2(0.68)(0.32)}{(0.02)^2} = 2089.8 \simeq 2090.$$

Therefore, if we base our estimate of p on a random sample of size 2090, we can be 95% confident that our sample proportion will not differ from the true proportion by more than 0.02.

Choice of Sample Size

Example 11

How large a sample is required if we want to be at least 95% confident that our estimate of p in Example 9 is within 0.02 of the true value?

Solution

Unlike in Example 10, we shall now assume that no preliminary sample has been taken to provide an estimate of p. Consequently, we can be at least 95% confident that our sample proportion will not differ from the true proportion by more than 0.02 if we choose a sample of size

$$n = \frac{(1.96)^2}{(4)(0.02)^2} = 2401.$$

Comparing the results of Examples 10 and 11, we see that information concerning p, provided by a preliminary sample or from experience, enables us to choose a smaller sample while maintaining our required degree of accuracy.

One-Sided Confidence Bounds

One-Sided Confidence Bounds

We may find approximate one-sided confidence bounds on p by a simple modification of Equation (22).

Theorem 15 (Approximate OneSided Confidence Bounds on a Binomial Proportion)

The approximate $100(1-\alpha)\%$ lower and upper confidence bounds are

$$\boxed{\widehat{p} - z_\alpha\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} < p \quad \text{and} \quad p < \widehat{p} + z_\alpha\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}} \tag{26}$$

respectively.